

ECE 315: Signals and Systems

Homework 2

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This homework is due on **2/13, 11:59pm**. For analytical problems, write your solutions independently by hand. For coding problems, attach your source code and answers. Combine them in one PDF document and submit on CANVAS. Please ensure that your PDF is clearly readable on a computer.

1. (2pts) Consider the following signals:

- $x(t) = \sin(\frac{\pi}{3}t)\delta(t)$
- $x(t) = \cos(\frac{\pi}{3}t)u(t)$
- $x(t) = 2u(t+1) - u(t) - u(t-1)$
- $x(t) = \int_{-\infty}^{\infty} e^{-\tau}(\delta(t-\tau) + \delta(t+\tau))d\tau$

Compute their energy and average power.

2. (2pts) Consider the following continuous-time systems with input $x(t)$ and output $y(t)$:

- $y(t) = e^{x(t)}$
- $y(t) = \frac{d}{dt}x(t-1)$
- $y(t) = (x(\frac{1}{2}t-1))^2$
- $y(t) = \int_{-\infty}^{\infty} x(\tau)(\delta(t-\tau) + \delta(t+\tau))d\tau$

Determine whether they are memoryless, invertible, causal, time-invariant, and linear. Justify your answer.

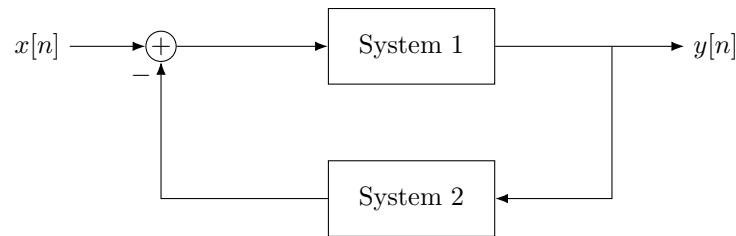
3. (2pts) Consider the following discrete-time systems with input $x[n]$ and output $y[n]$:

- $y[n] = x[n^2]$
- $y[n] = x[n] - x[1-n]$

- $y[n] = (x[2n - 1])^2$
- $y[n] = \sum_{k=-\infty}^{\infty} x[k](\delta[n - k] + \delta[n + k])$

Determine whether they are memoryless, invertible, causal, time-invariant, and linear. Justify your answer.

- (2pts) Prove that any time-invariant system with a periodic input has a periodic output. *Hint:* Consider a discrete-time time-invariant system with periodic input $x[n]$ with period N , show that the output $y[n]$ is periodic; repeat this exercise for a continuous-time time-invariant system.
- (2pts) Consider the following feedback system (w/ input $x[n]$ and output $y[n]$) that consists of two subsystems:
 - System 1 with input $x_1[n]$ and output $y_1[n]$: $y_1[n] = x_1[n] + x_1[n - 1]$.
 - System 2 with input $x_2[n]$ and output $y_2[n]$: $y_2[n] = x_2[n]$.



- Determine whether the feedback system is memoryless, invertible, causal, time-invariant, and linear. Justify your answer.
 - Assume that $y[n] = 0$ for $n < 0$. Sketch the outputs of the feedback system when $x[n] = \delta[n]$ and $x[n] = u[n]$.
- Plot the outputs of the feedback system in Question 5 when $x[n] = \delta[n]$ and $x[n] = u[n]$ using MATLAB or Python.